

EMILIO AGUINALDO COLLEGE

School of Physical Therapy

Functional Anatomy & Kinesiology — Post-Test Master Key

I. Hip & Pelvis Post-Test

1. Which structure forms the socket of the hip joint?

- A. Femoral head
- B. Iliac crest
- C. Acetabulum
- D. Pubic ramus

Correct Answer: C. Acetabulum

2. Which structure is primarily weight-bearing during sitting?

- A. Ischial tuberosity
- B. Pubic symphysis
- C. ASIS
- D. Iliac crest

Correct Answer: A. Ischial tuberosity

3. Which joint is classified as a synovial ball-and-socket joint?

- A. SI joint
- B. Pubic symphysis
- C. Lumbosacral joint
- D. Hip joint

Correct Answer: D. Hip joint (Marked as A on key, corrected to D)

4. Which structure deepens the acetabulum to improve stability?

- A. Bursa
- B. Joint capsule
- C. Ligamentum teres
- D. Labrum

Correct Answer: D. Labrum

5. Which ligament primarily prevents hyperextension of the hip?

- A. Iliofemoral ligament
- B. Pubofemoral ligament
- C. Ischiofemoral ligament
- D. Transverse ligament

Correct Answer: A. Iliofemoral ligament

6. Which ligament limits hip abduction and extension?

- A. Iliofemoral
- B. Pubofemoral
- C. Ischiofemoral
- D. Ligamentum teres

Correct Answer: B. Pubofemoral (Marked as C on key, corrected to B)

7. Which ligament limits internal rotation of the hip?

- A. Iliofemoral
- B. Pubofemoral
- C. Ischiofemoral
- D. Transverse ligament

Correct Answer: C. Ischiofemoral (Marked as B on key, corrected to C)

8. The normal angle of inclination of the femur in adults is approximately:

- A. 90°
- B. 110°
- C. 125°
- D. 145°

Correct Answer: C. 125°

9. A femoral neck-shaft angle less than 120° is classified as:

- A. Coxa valga
- B. Coxa vara
- C. Anteversion
- D. Retroversion

Correct Answer: B. Coxa vara

10. The normal angle of femoral anteversion in adults is:

- A. 0–5°
- B. 5–10°
- C. 10–15°
- D. 20–30°

Correct Answer: C. 10–15° (Marked as B on key, corrected to C)

11. A child with in-toeing gait MOST likely has:

- A. Coxa vara
- B. Coxa valga
- C. Decreased anteversion
- D. Increased anteversion

Correct Answer: D. Increased anteversion (Marked as C on key, corrected to D)

12. A patient with out-toeing gait MOST likely has:

- A. Increased anteversion
- B. Decreased anteversion
- C. Coxa vara
- D. Increased inclination

Correct Answer: B. Decreased anteversion (Marked as A on key, corrected to B)

13. The normal center-edge angle (Wiberg angle) is:

- A. 10–15°
- B. 15–20°
- C. 20–25°
- D. 25–40°

Correct Answer: D. 25–40° (Marked as C on key, corrected to D)

14. A decreased center-edge angle indicates:

- A. Increased stability
- B. Decreased stability
- C. Increased strength
- D. Increased ROM

Correct Answer: B. Decreased stability

15. Which structure is evaluated using Nélaton's line?

- A. Femoral head
- B. Lesser trochanter
- C. Greater trochanter
- D. Iliac crest

Correct Answer: C. Greater trochanter

16. Which artery is the primary blood supply to the femoral head?

- A. Medial femoral circumflex artery
- B. Lateral femoral circumflex artery
- C. Obturator artery
- D. Inferior gluteal artery

Correct Answer: A. Medial femoral circumflex artery (Marked as B on key, corrected to A)

17. Which artery primarily supplies the anterior femoral neck?

- A. Medial femoral circumflex artery
- B. Lateral femoral circumflex artery
- C. Obturator artery
- D. Superior gluteal artery

Correct Answer: B. Lateral femoral circumflex artery (Marked as A on key, corrected to B)

18. Which muscle group is MOST active during propulsion phase of gait?

- A. Hip flexors
- B. Hip abductors
- C. Hip extensors
- D. Hip internal rotators

Correct Answer: C. Hip extensors

19. Which muscle contributes to both hip flexion and knee extension?

- A. Sartorius
- B. Rectus femoris
- C. Iliopsoas
- D. Tensor fascia lata

Correct Answer: B. Rectus femoris (Marked as A on key, corrected to B)

20. Which muscle is the primary hip abductor?

- A. Gluteus maximus
- B. Gluteus minimus
- C. Gluteus medius
- D. Iliopsoas

Correct Answer: C. Gluteus medius (Marked as A on key, corrected to C)

21. Injury to which nerve causes weakness of hip abductors?

- A. Femoral nerve
- B. Obturator nerve
- C. Sciatic nerve
- D. Superior gluteal nerve

Correct Answer: D. Superior gluteal nerve

22. Which nerve innervates most hip adductors?

- A. Femoral nerve
- B. Obturator nerve
- C. Sciatic nerve
- D. Inferior gluteal nerve

Correct Answer: B. Obturator nerve (Marked as C on key, corrected to B)

23. Which muscle assists in hip internal rotation?

- A. Piriformis
- B. Obturator externus
- C. Quadratus femoris
- D. Gluteus minimus

Correct Answer: D. Gluteus minimus (Marked as A on key, corrected to D)

24. The normal range of hip flexion (knee flexed) is:

- A. 0–90°
- B. 0–100°
- C. 0–120°
- D. 0–140°

Correct Answer: C. 0–120°

25. The normal range of hip extension is:

- A. 0–10°
- B. 0–15°
- C. 0–20°
- D. 0–30°

Correct Answer: C. 0–20° (Marked as B on key, corrected to C)

26. The normal range of hip abduction is:

- A. 0–30°
- B. 0–45°
- C. 0–60°
- D. 0–75°

Correct Answer: B. 0–45°

27. The normal range of hip internal rotation is:

- A. 0–20°
- B. 0–25°
- C. 0–35°
- D. 0–45° / 0–50°

Correct Answer: D. 0–45° / 0–50° (Marked as C on key, corrected to D)

28. Which motion occurs in the frontal plane?

- A. Flexion
- B. Extension
- C. Abduction
- D. Internal rotation

Correct Answer: C. Abduction

29. Which plane involves internal and external rotation?

- A. Sagittal
- B. Frontal
- C. Coronal
- D. Transverse

Correct Answer: D. Transverse (Marked as A on key, corrected to D)

30. During hip flexion, the arthrokinematics is:

- A. Posterior roll, anterior glide
- B. Anterior roll, posterior glide
- C. Superior roll, inferior glide
- D. Inferior roll, superior glide

Correct Answer: A. Posterior roll, spin/anterior glide (Marked as A on key)

31. Which motion involves superior roll and inferior glide?

- A. Flexion
- B. Extension
- C. Abduction
- D. Adduction

Correct Answer: C. Abduction (Marked as A on key, corrected to C)

32. Which movement increases lumbar lordosis?

- A. Posterior pelvic tilt
- B. Lateral tilt
- C. Anterior pelvic tilt
- D. Pelvic rotation

Correct Answer: C. Anterior pelvic tilt

33. Which pelvic motion occurs when one iliac crest moves superiorly?

- A. Lateral tilt
- B. Pelvic rotation
- C. Posterior tilt
- D. Anterior tilt

Correct Answer: A. Lateral tilt

34. Which joint has the least mobility but greatest stability?

- A. Hip joint
- B. SI joint
- C. Pubic symphysis
- D. Knee joint

Correct Answer: C. Pubic symphysis (Marked as B on key, corrected to C)

35. Which function of the hip is MOST related to absorbing forces during gait?

- A. Mobility
- B. Stability
- C. Coordination
- D. Shock absorption

Correct Answer: D. Shock absorption

II. Knee Post-Test

1. A physical therapist is assessing a patient with difficulty extending the knee against resistance. Which of the following muscles is considered the primary and most powerful contributor to knee extension?

- a. Biceps femoris
- b. Semitendinosus
- c. Semimembranosus
- d. Adductor magnus
- e. None of the above

Correct Answer: e. None of the above (The primary extension group is the Quadriceps femoris)

2. The patella plays several important biomechanical roles in the knee joint. Which of the following statements does NOT correctly describe one of its functions?

- I. Improves efficiency of the quadriceps by increasing the moment arm
 - II. Contributes to knee joint stability
 - III. Increases friction and compression during movement
 - IV. Protects the femoral condyles from trauma during flexion
- a. All are correct
 - b. I, II, III
 - c. II and IV
 - d. I and III
 - e. Only III

Correct Answer: e. Only III (The patella decreases friction, not increases it; statement III is incorrect)

3. During a clinical ligamentous stress test, a therapist observes excessive anterior displacement of the tibia relative to the femur. This finding most strongly suggests injury to which structure?

- a. Posterior cruciate ligament
- b. Lateral collateral ligament
- c. Medial collateral ligament
- d. Anterior cruciate ligament

Correct Answer: d. Anterior cruciate ligament

4. A clinician is measuring the Q-angle of a patient to assess patellofemoral alignment. Which of the following statements is NOT correct regarding the Q-angle?

- a. It is formed by the intersection of the quadriceps tendon and patellar ligament
- b. Increased values may predispose to patellar maltracking
- c. A larger Q-angle may lead to patellar subluxation
- d. One line is drawn from the patella to the ASIS
- e. One line is drawn from the patella to the tibial tubercle

Correct Answer: a. It is formed by the intersection of the quadriceps tendon and patellar ligament (It is formed by lines from ASIS to mid-patella, and mid-patella to tibial tubercle)

5. Which of the following statements about the anterior cruciate ligament (ACL) is incorrect?

- a. It attaches to the lateral femoral condyle
- b. It runs in a superior, posterior, and lateral direction
- c. It attaches to the anterior horn of the medial meniscus
- d. It has an oblique orientation within the knee joint
- e. None of the above

Correct Answer: c. It attaches to the anterior horn of the medial meniscus (Marked as B/C on sheet)

6. A patient presents after a valgus injury to the knee while the foot was planted. Which of the following statements regarding the collateral ligaments is NOT accurate?

- a. The lateral collateral ligament connects the femur to the fibula
- b. The medial collateral ligament connects the femur to the tibia
- c. These ligaments resist varus and valgus stresses
- d. They provide stability during knee extension
- e. They are only injured with rotational forces

Correct Answer: e. They are only injured with rotational forces

7. Regarding the medial meniscus, which of the following statements is NOT true?

- a. It is crescent-shaped and located medially
- b. It is completely avascular
- c. It is stabilized by coronary ligaments
- d. It is commonly involved in the "unhappy triad"
- e. All are correct

Correct Answer: b. It is completely avascular (The outer zone has vascularity)

8. The cruciate ligaments are unique in their anatomical position within the knee. Which combination correctly describes their location?

- I. Intracapsular
- II. Extracapsular
- III. Intrasynovial
- IV. Extrasynovial
- a. I and III
- b. II and IV
- c. I and IV
- d. II and III

Correct Answer: c. I and IV (Intracapsular, but extrasynovial)

9. The screw-home mechanism is an important component of terminal knee extension. Which of the following statements is NOT correct?

- I. It is purely a mechanical process
- II. It cannot be voluntarily controlled
- III. The popliteus muscle unlocks the knee
- IV. It occurs during the last 20° of extension
- a. All are correct
- b. I, II, III
- c. I and III
- d. II and IV
- e. Only IV

Correct Answer: a. All are correct (Bonus noted on key)

10. The menisci contribute to several functions within the knee joint. Which of the following is NOT one of their functions?

- I. Joint lubrication
- II. Allowing capsular intrusion
- III. Enhancing joint stability
- IV. Reducing contact area
- a. I, II, III
- b. I and III
- c. II and IV
- d. Only IV
- e. None

Correct Answer: c. II and IV (Menisci increase contact area to spread load, and prevent intrusion)

11. During knee flexion, how do the menisci typically move relative to the tibial plateau?

- a. Anteriorly, medial more than lateral
- b. Posteriorly, medial more than lateral
- c. Anteriorly, lateral more than medial
- d. Posteriorly, lateral more than medial
- e. Equal posterior displacement

Correct Answer: d. Posteriorly, lateral more than medial

12. In a closed-chain activity such as squatting, what is the typical arthrokinematic pattern observed during knee flexion?

- a. Rolling predominates early, sliding later
- b. Pure rolling occurs
- c. Pure sliding occurs
- d. Sliding predominates early, rolling later
- e. Equal rolling and sliding

Correct Answer: a. Rolling predominates early, sliding later

13. A ligament that connects the posterior horn of the lateral meniscus to the femur and lies anterior to the PCL is known as:

- a. Coronary ligament
- b. Cotyloid ligament
- c. Poupart's ligament
- d. Ligament of Humphrey
- e. Ligament of Wrisberg

Correct Answer: d. Ligament of Humphrey (Anterior to PCL is Humphrey; Posterior is Wrisberg)

14. The anterior cruciate ligament follows a specific directional orientation within the knee joint. Which of the following best describes this orientation?

- a. Upward, backward, and lateralward
- b. Upward, forward, and medialward
- c. Upward, forward, and lateralward
- d. Downward, backward, and medialward
- e. Downward, forward, and lateralward

Correct Answer: a. Upward, backward, and lateralward (from tibia to femur)

15. During closed-chain motion such as weight-bearing, the posterior cruciate ligament primarily functions to prevent:

- a. Anterior displacement of the femur
- b. Posterior displacement of the femur
- c. Anterior displacement of the tibia
- d. Posterior displacement of the tibia
- e. None of the above

Correct Answer: a. Anterior displacement of the femur (or posterior displacement of tibia relative to femur)

BONUS 16. A student describes knee arthrokinematics during extension. Which statement is most accurate?

- a. Both statements are correct
- b. Both statements are incorrect
- c. First statement is correct, second incorrect
- d. First statement incorrect, second correct

Correct Answer: Marked as Bonus

17. Which muscle is considered the primary dynamic stabilizer of the patella during knee extension?

- a. Rectus femoris
- b. Vastus lateralis
- c. Vastus intermedius
- d. Vastus medialis longus
- e. Vastus medialis oblique

Correct Answer: e. Vastus medialis oblique

18. During knee flexion, at approximately what angle does the patellofemoral joint experience the greatest contact area?

- a. Full extension
- b. Full flexion
- c. 25°
- d. 90°
- e. 135°

Correct Answer: d. 90° (Contact area maxes out around 60–90°)

19. A patient presents with anterior knee pain due to abnormal patellar tracking. Which structural abnormality is most likely responsible?

- a. Patella alta
- b. Patella baja
- c. Both
- d. None

Correct Answer: a. Patella alta

20. Which of the following conditions may result in weakness or inability to actively extend the knee?

- a. Patella alta
- b. Patella baja
- c. Both
- d. None

Correct Answer: a. Patella alta (Marked as A/B on key)

21. Lateral patellar dislocation is commonly associated with which biomechanical factors?

- I. Genu valgum
 - II. Increased femoral anteversion
 - III. Patella alta
 - IV. Genu varum
- a. I, II, III
 - b. I and III
 - c. All
 - d. Only IV
 - e. I and IV

Correct Answer: a. I, II, III

22. In rehabilitation following ACL injury, strengthening which muscle group is most important to reduce anterior tibial translation?

- a. Quadriceps
- b. Hamstrings
- c. Gastrocnemius
- d. Gluteus maximus

Correct Answer: b. Hamstrings (Pull the tibia posteriorly)

23. A patient involved in a motor vehicle accident sustains a posteriorly directed force to the tibia while the knee is flexed. This mechanism is most commonly associated with injury to which ligament?

- a. ACL
- b. PCL
- c. MCL
- d. LCL

Correct Answer: b. PCL ("Dashboard injury")

24. A blow to the lateral side of the knee while the foot is planted is most likely to injure which structure?

- a. ACL
- b. PCL
- c. MCL
- d. LCL

Correct Answer: c. MCL (Valgus stress)

25. The normal Q-angle in males is approximately:

- a. 13°
- b. 20°
- c. 5°
- d. 25°
- e. None

Correct Answer: a. 13° (Normal range for males is 10–14°)

26. A patient presents with a knee alignment where the medial angle is less than 170°. Which deformity does this indicate?

- a. Genu varum
- b. Genu valgum
- c. Genu recurvatum
- d. Both A & B
- e. None

Correct Answer: b. Genu valgum (<165-170° is knock-knees / valgus)

27. How many muscles that act on the knee are classified as one-joint muscles?

- a. 2
- b. 3
- c. 4
- d. 5
- e. 6

Correct Answer: b. 3 (Vastus lateralis, vastus medialis, vastus intermedius)

28. Among the quadriceps group, which muscle requires the least force to produce knee extension?

- a. Vastus medialis oblique
- b. Vastus lateralis
- c. Vastus medialis longus
- d. Vastus intermedius
- e. Rectus femoris

Correct Answer: c. Vastus medialis longus (Marked as C on sheet)

29. Which muscle group assists the posterior cruciate ligament in preventing posterior translation of the tibia?

- a. Hamstrings
- b. Quadriceps
- c. Popliteus
- d. Sartorius
- e. Gracilis

Correct Answer: b. Quadriceps (Quadriceps acts as a PCL agonist)

30. Which statement regarding the Q-angle is incorrect?

- a. Formed by quadriceps muscle alignment
- b. Increased angle may cause lateral tracking
- c. Can contribute to patellar instability
- d. Measured using ASIS and tibial tubercle
- e. Involves the triceps femoris

Correct Answer: e. Involves the triceps femoris

31. A ligament injury caused by varus stress is most likely affecting which structure?

- a. LCL
- b. MCL
- c. ACL
- d. PCL

Correct Answer: a. LCL

32. The oblique popliteal ligament originates primarily from which muscle?

- a. Semitendinosus
- b. Semimembranosus
- c. Gracilis
- d. Biceps femoris

Correct Answer: b. Semimembranosus

33. Which statement about the lateral meniscus is incorrect?

- a. It is more circular in shape
- b. It is more mobile than medial meniscus
- c. It is more frequently injured
- d. It is deeper
- e. It has fewer attachments

Correct Answer: c. It is more frequently injured (The medial meniscus is more static and frequently torn)

34. During the Lachman test, the knee should ideally be positioned in:

- a. 20–30° flexion
- b. 30–40° flexion
- c. 40–50° flexion
- d. 80–90° flexion

Correct Answer: a. 20–30° flexion

35. To assess the integrity of the medial collateral ligament, which force is applied?

- a. Valgus stress
- b. Varus stress
- c. Posterior force
- d. Anterior force

Correct Answer: a. Valgus stress

36. Which statement about the rectus femoris is incorrect?

- a. It is a two-joint muscle
- b. It originates from the AIIS
- c. It assists in hip flexion
- d. It is not palpable proximally
- e. It extends the knee

Correct Answer: d. It is not palpable proximally

37. The lateral collateral ligament can be palpated between which two anatomical landmarks?

- a. Lateral epicondyle and fibular head
- b. Medial epicondyle and tibia
- c. Patella and tibial tubercle
- d. None

Correct Answer: a. Lateral epicondyle and fibular head

38. The "unhappy triad" of the knee does NOT include:

- a. Medial meniscus
- b. ACL
- c. MCL
- d. None

Correct Answer: d. None (Includes ACL, MCL, and Medial Meniscus)

39. In which position is the posterior cruciate ligament relatively slack?

- a. Full extension
- b. 20–30° flexion
- c. 30–60° flexion
- d. None

Correct Answer: b. 20–30° flexion (Marked as A on key, corrected to B)

40. Which joint is considered the largest synovial joint in the human body?

- a. Knee joint
- b. Hip joint
- c. Shoulder joint
- d. Sacroiliac joint

Correct Answer: a. Knee joint

III. Leg, Ankle, & Foot Post-Test

1. A patient presents with recurrent episodes of ankle instability following an inversion injury sustained during landing. Imaging reveals damage to the ligament most vulnerable when the foot is positioned in plantarflexion combined with inversion. Which structure is MOST likely injured?

- a. Deltoid ligament
- b. Posterior talofibular ligament
- c. Calcaneofibular ligament
- d. Anterior talofibular ligament
- e. Tibionavicular ligament

Correct Answer: d. Anterior talofibular ligament (ATFL)

2. During closed kinetic chain pronation, which triplanar combination correctly occurs at the foot?

- a. Inversion, abduction, dorsiflexion
- b. Eversion, adduction, plantarflexion
- c. Inversion, adduction, plantarflexion
- d. Eversion, abduction, dorsiflexion
- e. Inversion, abduction, dorsiflexion

Correct Answer: b. Eversion, adduction, plantarflexion (CKC pronation involves calcaneal eversion with talar plantarflexion and adduction)

3. Which bone serves as the keystone of the medial longitudinal arch and is considered the MOST clinically significant landmark during navicular drop assessment?

- a. Cuboid
- b. Talus
- c. Navicular
- d. Sustentaculum tali
- e. Second cuneiform

Correct Answer: c. Navicular

4. A mobilization technique intended to improve ankle dorsiflexion should emphasize which arthrokinematic glide of the talus?

- a. Anterior glide
- b. Medial glide
- c. Lateral glide
- d. Posterior glide
- e. Superior glide

Correct Answer: d. Posterior glide

5. Which statement BEST explains why the talus is highly susceptible to avascular necrosis after fracture?

- a. It bears minimal body weight
- b. It has extensive muscular attachments
- c. It possesses limited vascular supply and extensive cartilage coverage
- d. It lacks ligamentous support
- e. It articulates only with the calcaneus

Correct Answer: c. It possesses limited vascular supply and extensive cartilage coverage

6. During gait, the midtarsal joint becomes rigid primarily during which functional state?

- a. Pronation
- b. Eversion
- c. Supination
- d. Dorsiflexion
- e. Weight acceptance

Correct Answer: c. Supination (Locks the midtarsal joint for push-off)

7. Which muscle contributes MOST significantly to maintenance of the medial longitudinal arch dynamically?

- a. Fibularis brevis
- b. Tibialis posterior
- c. Flexor hallucis longus
- d. Gastrocnemius
- e. Tibialis anterior

Correct Answer: b. Tibialis posterior

8. A patient demonstrates foot slap immediately after heel strike. Weakness of which muscle is MOST likely responsible?

- a. Gastrocnemius
- b. Fibularis longus
- c. Tibialis anterior
- d. Soleus
- e. Tibialis posterior

Correct Answer: c. Tibialis anterior

9. Which structure acts as the primary passive restraint against excessive eversion?

- a. Calcaneofibular ligament
- b. Anterior talofibular ligament
- c. Posterior talofibular ligament
- d. Deltoid ligament
- e. Plantar fascia

Correct Answer: d. Deltoid ligament

10. Which metatarsal is considered the MOST rigid due to its articulation with the cuneiforms and acts as the keystone of forefoot stability?

- a. First metatarsal
- b. Second metatarsal
- c. Third metatarsal
- d. Fourth metatarsal
- e. Fifth metatarsal

Correct Answer: b. Second metatarsal

11. A patient with pes cavus would MOST likely demonstrate which biomechanical characteristic?

- a. Excessive pronation
- b. Increased shock absorption
- c. Flexible hypermobile foot
- d. Rigid supinated foot
- e. Collapsed transverse arch

Correct Answer: d. Rigid supinated foot

12. Which ligament directly supports the head of the talus in the medial longitudinal arch?

- a. Long plantar ligament
- b. Plantar fascia
- c. Spring ligament (Plantar calcaneonavicular)
- d. Deltoid ligament
- e. Interosseous talocalcaneal ligament

Correct Answer: c. Spring ligament

13. Which component is unique to the subtalar joint motion?

- a. Pure sagittal plane movement
- b. Uniplanar rotation only
- c. Complex triplanar movement around an oblique axis
- d. Translatory movement without rotation
- e. Motion restricted to inversion only

Correct Answer: c. Complex triplanar movement around an oblique axis

14. The fibula contributes minimally to weight-bearing, yet fractures significantly impair ankle function primarily because the fibula:

- a. Serves as the attachment for the plantar fascia
- b. Forms the subtalar joint
- c. Maintains ankle mortise integrity
- d. Contains the dorsalis pedis artery
- e. Prevents tibial torsion

Correct Answer: c. Maintains ankle mortise integrity

15. Which intrinsic foot muscle is the ONLY muscle located on the dorsum of the foot?

- a. Extensor hallucis brevis / Extensor digitorum brevis
- b. Extensor digitorum brevis
- c. Dorsal interossei
- d. Flexor digitorum brevis
- e. Quadratus plantae

Correct Answer: b. Extensor digitorum brevis (and Extensor hallucis brevis)

16. A positive navicular drop test is MOST associated with which condition?

- a. Pes cavus
- b. Pes varus
- c. Hallux rigidus
- d. Pes planus
- e. Morton's foot

Correct Answer: d. Pes planus

17. Which artery is palpated between the first and second metatarsals on the dorsum of the foot?

- a. Posterior tibial artery
- b. Fibular artery
- c. Plantar arch
- d. Dorsalis pedis artery
- e. Medial plantar artery

Correct Answer: d. Dorsalis pedis artery

18. In open kinetic chain supination, which combination correctly occurs?

- a. Eversion, abduction, plantarflexion
- b. Inversion, adduction, plantarflexion
- c. Inversion, abduction, dorsiflexion
- d. Eversion, abduction, dorsiflexion
- e. Eversion, inversion, plantarflexion

Correct Answer: b. Inversion, adduction, plantarflexion

19. Which structure within the sinus tarsi contributes MOST to proprioceptive feedback and subtalar stability?

- a. Spring ligament
- b. Deltoid ligament
- c. Interosseous talocalcaneal ligament
- d. Calcaneofibular ligament
- e. Long plantar ligament

Correct Answer: c. Interosseous talocalcaneal ligament

20. A patient presents with numbness and tingling along the plantar aspect of the foot due to compression beneath the flexor retinaculum. Which condition is MOST consistent with these findings?

- a. Sinus tarsi syndrome
- b. Morton neuroma
- c. Tarsal tunnel syndrome
- d. High ankle sprain
- e. Plantar fasciitis

Correct Answer: c. Tarsal tunnel syndrome

21. Which muscle is primarily responsible for postural control due to its predominance of slow-twitch fibers?

- a. Gastrocnemius
- b. Tibialis posterior
- c. Fibularis longus
- d. Soleus
- e. Flexor hallucis longus

Correct Answer: d. Soleus

22. Which foot type pattern demonstrates the second toe longer than the hallux and is associated with increased stress on the second metatarsal?

- a. Roman foot
- b. Egyptian foot
- c. Morton's foot
- d. Splay foot
- e. Pes cavus

Correct Answer: c. Morton's foot

23. During dorsiflexion at the talocrural joint, the talus demonstrates which arthrokinematic motion?

- a. Posterior roll and anterior glide
- b. Anterior roll and posterior glide
- c. Medial roll and lateral glide
- d. Posterior roll and posterior glide
- e. Anterior roll and anterior glide

Correct Answer: b. Anterior roll and posterior glide

24. Which muscle group contributes approximately 80–85% of propulsive force during gait?

- a. Dorsiflexors
- b. Evertors
- c. Invertors
- d. Plantarflexors
- e. Intrinsic foot muscles

Correct Answer: d. Plantarflexors

25. A syndesmotic injury primarily involves which articulation?

- a. Talonavicular joint
- b. Subtalar joint
- c. Distal tibiofibular joint
- d. Calcaneocuboid joint
- e. Talocalcaneal joint

Correct Answer: c. Distal tibiofibular joint ("High ankle sprain")

26. Which structure forms the keystone of the transverse arch?

- a. Navicular
- b. Cuboid
- c. Talus
- d. Second cuneiform
- e. First metatarsal

Correct Answer: d. Second cuneiform

27. Weakness of the intrinsic muscles leading to widening of the forefoot and collapse of the transverse arch describes which condition?

- a. Pes valgus
- b. Splay foot
- c. Hallux valgus
- d. Pes cavus
- e. Morton neuroma

Correct Answer: b. Splay foot

28. Which muscle functions both as a strong inverter and primary dorsiflexor of the ankle?

- a. Tibialis posterior
- b. Fibularis brevis
- c. Tibialis anterior
- d. Gastrocnemius
- e. Soleus

Correct Answer: c. Tibialis anterior

29. Which ligament of the lateral ankle is LEAST commonly injured due to its relative strength and position?

- a. Anterior talofibular ligament
- b. Calcaneofibular ligament
- c. Posterior talofibular ligament
- d. Deltoid ligament
- e. Tibionavicular ligament

Correct Answer: c. Posterior talofibular ligament (PTFL is the strongest/least injured of the lateral trio)

30. Which statement BEST describes the biomechanical behavior of the talocrural joint during plantarflexion?

- a. Increased congruency due to wider anterior talus
- b. Maximum stability because of mortise compression
- c. Decreased stability because the narrower posterior talus occupies the mortise
- d. Complete locking of the ankle complex
- e. Elimination of translatory motion

Correct Answer: c. Decreased stability because the narrower posterior talus occupies the mortise